

Fast Substrings Search Using Suffix Arrays

Data Structures and Algorithms – 3 (25CS2103E)

Problem Statement 13 | Team No. 27

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Abstract

This project presents a high-speed substring lookup system using Suffix Arrays for efficient text searching. The system preprocesses the input text by constructing and sorting its suffixes into a suffix array, enabling multiple substring queries to be answered without repeatedly scanning the entire text. For each query, binary search is used to quickly identify matching occurrences and return their positions in the original text. The system also records query execution time to evaluate search performance. It is designed to handle large text files and Unicode characters, demonstrating how efficient preprocessing and advanced data structures can improve repeated substring searching compared with conventional approaches.

PROBLEM STATEMENT

The Core Challenge

Repeatedly searching for substrings in a large text can become inefficient when every query requires scanning the full text again. As the text grows and the number of queries increases, a more efficient approach is needed.

- Preprocess the input text using a suffix array
- Support multiple substring queries
- Avoid repeatedly scanning the complete text
- Return occurrence positions for matching substrings
- Measure query execution time
- Support large text files and Unicode characters

Proposed Functionalities

1. Input Text
2. Build Suffix Array
3. Sort/Preprocess
4. Enter Query
5. Binary Search
6. Show Results

System Features

- Large text input support
- Suffix array construction and sorted indexing
- Binary-search-based substring lookup with occurrence positions
- Query execution-time recording and Unicode support

Tools and Technologies

Programming Language

- Java

Core DSA Concepts

- Suffix Array
- Binary Search
- Sorting
- Arrays & Strings

Frontend

- HTML
- CSS
- JavaScript

Development Tools

- VS Code
- Git / GitHub

Expected Outcome

- Functional System
- Efficient Repeated Searching
- Occurrence Position Retrieval
- Query Execution-Time Measurement
- Large Text and Unicode Support
- Advanced DSA Demonstration